

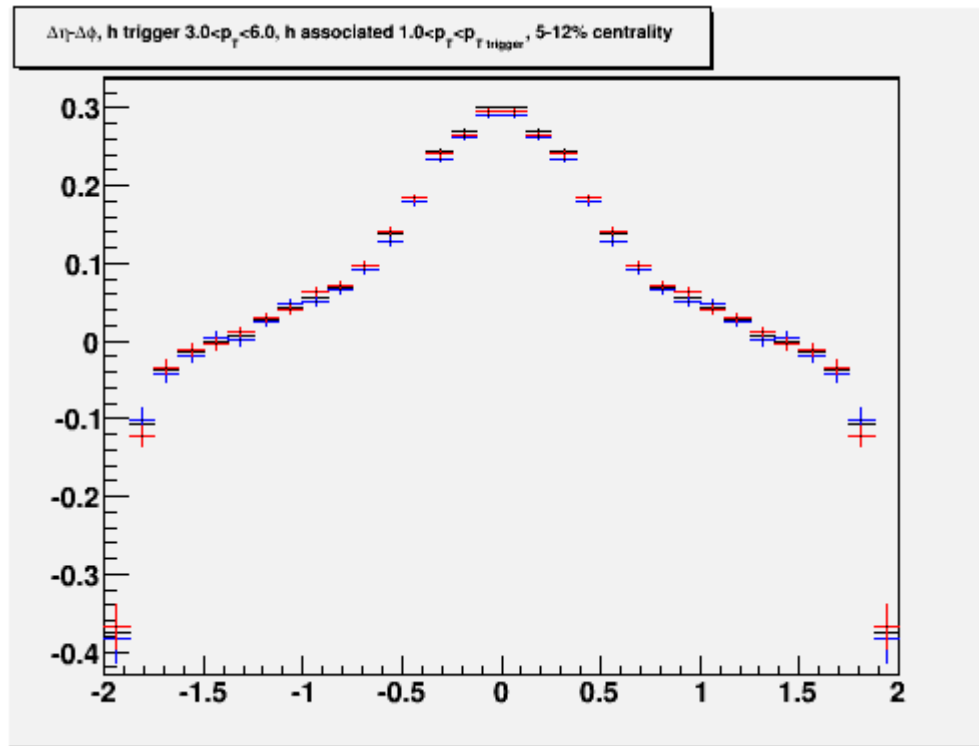
Collision system and energy dependence of di-hadron correlations

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Notes

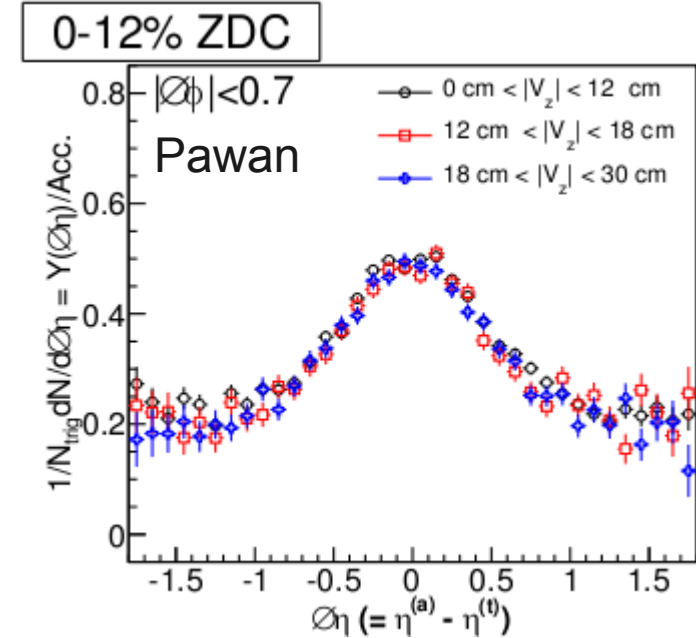
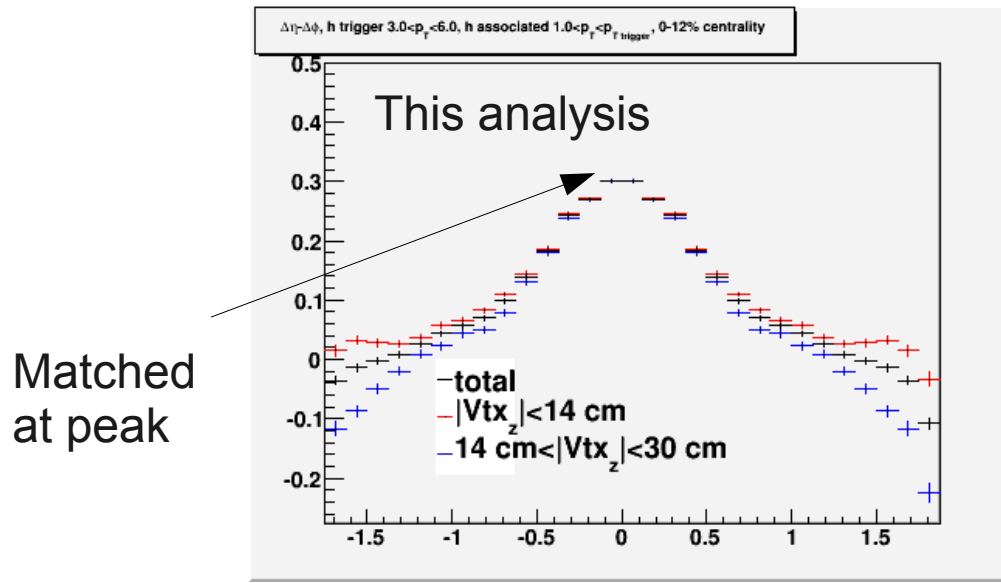
- We *know* the track reconstruction efficiency is greater for a smaller vertex z and we use the vertex z averaged efficiency. (This is taken care of in the analysis by using embedding with a realistic vertex z distribution.) This means the overall scale of the histograms should not be exactly the same. Judging by the histograms without background subtraction, the difference is about 5% from the smallest vertex z bin to the largest. I had played around with different scalings to correct for this but we don't *know* the true scale so the plots in the following pages are not rescaled
- Redoing the entire analysis with a tighter vertex z cut for the Au+Au 200 is not feasible. The efficiency would need to be determined again and both the standard analysis and the track merging correction would need to be done again. This would take several months.

Good news



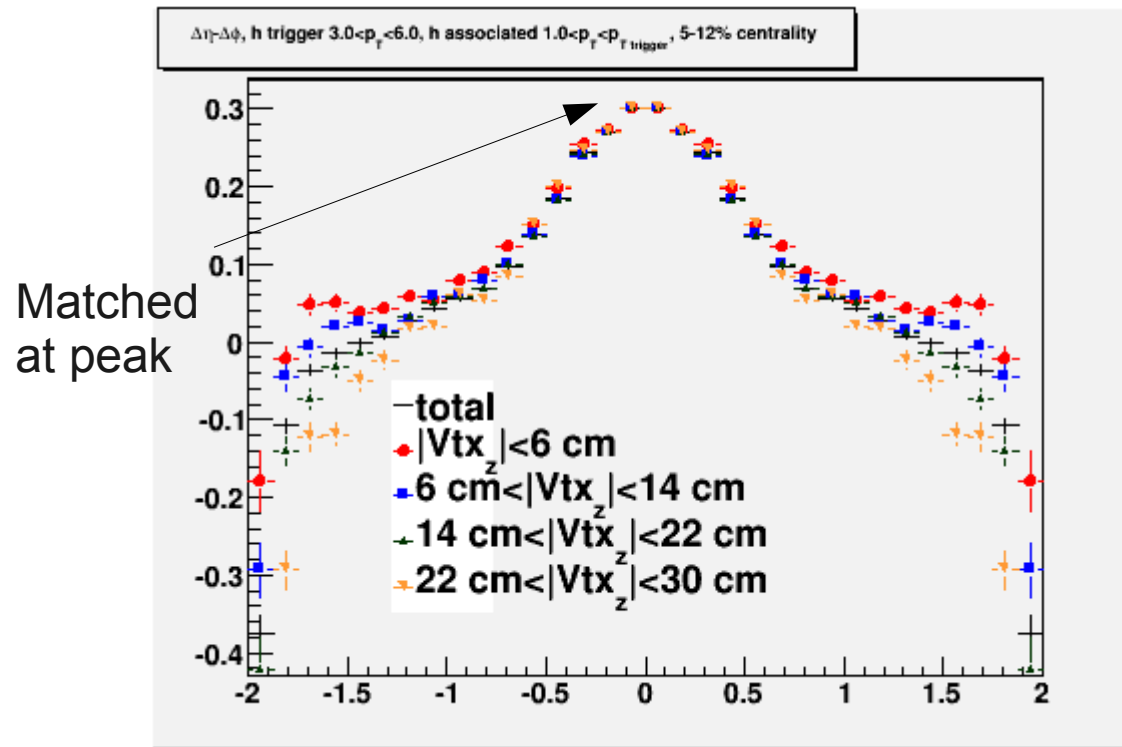
- Black = total, red = $2 < v_{TXz} < 30$, blue = $-30 < v_{TXz} < -2$
- No difference between positive and negative v_{TXz}

Vtx z dependence



- Difference between low and high z at ~ 1.7 :
 - Pawan: ~ 0.075 (with larger statistical errors)
 - This analysis: ~ 0.1
 - Slight difference in vtx z binning in the right direction to explain the difference between the two
- Effect is *not* eliminated by true mixed events

Vtx z binning: finer bins



- Width:
 - Difference between smallest vtx z bin and total: 5%
 - Had already proposed 4% systematic error, mainly due to fit range, in latest iteration
 - Difference in width between d+Au and central Au+Au 200 in paper draft: 60%
 - Physics claims are robust
- Yield:
 - Difference between smallest vtx z bin and total: 20%+5% due to tracking efficiency with vtx z
 - Almost definitely a pT-dependent statement: this is one of the worst kinematic ranges in the paper for such effects
 - Physics point in the paper is that there is no difference in yields

Possible physics vtx z dependence?

- Suppose there were a large difference in the jet-like correlation width for pairs with a mean $\eta=0$ and $\eta=1$ (but still with $d\eta\sim 0$)
- Events with vtx $z=0$ will have a much lower acceptance for track pairs with a mean $\eta=1$ than events with vtx $z=30$.
- This could lead to a difference in the jets sampled by the correlations
- But this is also completely untestable in this analysis...

Alternate plans

- Do full mixed events:
 - Au+Au 200 0-12% mixed events are 55% done now.
 - Rate $\sim 30\%$ /week $\rightarrow$ a little over 1 week left (3.3 weeks total)
 - Au+Au 200 min bias mixed events would take slightly less time, but also need to do Cu+Cu 62, Cu+Cu 200, Au+Au 62, and d+Au 200 data sets
 - Approximately 6 weeks to run over all remaining data, plus 1-2 days code development, remaking plots, and updating analysis note
 - **But still would not “fix” the issue**
- Apply systematic errors:
 - Redo the exercise on slide 5 for a few different pT cuts
 - $1.5 < pT_{\text{assoc}} < pT_{\text{trig}}$, $3 < pT_{\text{trig}} < 6$ GeV/c (standard cuts)
 - $1.5 < pT_{\text{assoc}} < pT_{\text{trig}}$, $2 < pT_{\text{trig}} < 2.5$ GeV/c (lowest pTtrig)
 - (Pawan's cuts are essentially lowest pTassoc)
 - Determine pT dependent systematic errors
 - May need to re-evaluate errors on slopes
 - Time scale 1 week, mainly due to lack of time to actually do this